

Digital Material Passport

ID DMP-METAL-EN10029-001 - Version 1.0.0  
Issue Date 2025-05-20 - Certificate Type EN 10204 3.1

| Manufacturer                                                                                                                | Customer                                                                                                            | Business Transaction |                                                 |
|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------|-------------------------------------------------|
| ThyssenKrupp Steel Europe AG<br>Kaiser-Wilhelm-Straße 100<br>47166 Duisburg<br>DE<br>quality@thyssenkrupp-steel.example.com | Heavy Industries Construction AB<br>Industrivägen 25<br>Göteborg<br>41705<br>SE<br>procurement@heavy-ind.example.se | Order                | PO-2025-1547 · 24800 kg<br>Pos. 10 · 2025-04-10 |
|                                                                                                                             |                                                                                                                     | Delivery             | DN-2025-8854 · 24800 kg<br>Pos. 10 · 2025-05-20 |

Product

|                    |                                            |                 |                                                          |
|--------------------|--------------------------------------------|-----------------|----------------------------------------------------------|
| Product Name       | Hot-rolled structural steel plate S355J2+N | Number (EN)     | 1.0577                                                   |
| Batch ID           | H-58742-12                                 | Name (Material) | S355J2+N                                                 |
| Surface Condition  | Hot-rolled                                 | Shape           | Plate - Width 2000 mm - Thickness 25 mm - Length 4500 mm |
| Delivery Condition | +N - Normalized                            | Product Norm    | EN 10025-2 (2019)                                        |
| Weight             | 24800 kg                                   | Product Norm    | EN 10029 (2010)                                          |
| Production Date    | 2025-05-18                                 |                 |                                                          |
| Country of Origin  | DE                                         |                 |                                                          |
| Tolerance Standard | EN 10029:2010                              |                 |                                                          |

Chemical Analysis

Heat Number H-58742 - Melting Process BOF+LF - Casting Method ContinuousCasting - Casting Date 2025-05-17 - Sample Location Ladle

| Symbol | C    | Mn   | Si   | P     | S     |
|--------|------|------|------|-------|-------|
| Unit   | %    | %    | %    | %     | %     |
| Min    |      |      |      |       |       |
| Max    | 0.20 | 1.60 | 0.50 | 0.025 | 0.020 |
| Actual | 0.18 | 1.52 | 0.28 | 0.015 | 0.009 |

Mechanical Properties

| Property                                                     | Symbol | Actual  | Minimum | Maximum | Method        |
|--------------------------------------------------------------|--------|---------|---------|---------|---------------|
| Tensile Strength <sup>1</sup>                                | Rm     | 515 MPa | 470 MPa | 630 MPa | EN ISO 6892-1 |
| Yield Strength <sup>1</sup>                                  | ReH    | 378 MPa | 355 MPa |         | EN ISO 6892-1 |
| Elongation after fracture<br>At room temperature, L0=5.65√S0 | A      | 24 %    | 20 %    |         | EN ISO 6892-1 |
| Charpy V-notch Impact Energy<br>At -20°C, longitudinal       | KV     | 45 J    | 27 J    |         | EN ISO 148-1  |

<sup>1</sup> At room temperature

Supplementary Tests

| Test                                                        | Result / limits                     | Method   | Status |
|-------------------------------------------------------------|-------------------------------------|----------|--------|
| Thickness<br>Measured at 4 points                           | 24.8 mm - min 24.7 mm - max 26.3 mm | EN 10029 | ✓      |
| Flatness Deviation<br>Measuring length 2000mm, Steel type L | 7 mm - max 10 mm                    | EN 10029 | ✓      |
| Edge Camber<br>Measured over full length                    | 8.00 mm - max 11.25 mm              | EN 10029 | ✓      |
| Out-of-Squareness<br>Measured at corners                    | 15 mm - max 20 mm                   | EN 10029 | ✓      |

Validation

We hereby certify that all material described above has been manufactured and tested in accordance with the requirements of EN 10025-2:2019, EN 10029:2010, and EN 10204:2004 type 3.1. The results comply with the requirements for S355J2+N steel grade and dimensional tolerances class B-S-G.

Validated by Dr. Werner Schmidt, Head of Quality Control, Quality Assurance - 2025-05-20